

L10b: The NPV and Probability of Profit of Composite Contracts

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Objectives for Today

First, two examples on the Greeks, the POE, and the POP. Then: what is a composite contract worth, and when does it make money?

Today's concepts

- The POE and POP of a single contract: from simulated share prices, compared with Delta.
- The NPV and POP of a composite contract: sums over its legs.
- Common composite contracts: outlook, breakevens, maximum profit and loss, and why closing early earns less.

What You Need to Know

Core concepts

- POE: the probability a contract finishes in the money. POP: the probability its profit at expiration is positive. Both come from simulated share prices, and the POE $\sim |\Delta_\star|$.
- Profit at expiration: the legs' payoffs, weighted by $\theta_i n_i$, minus the composite premium grown to expiration. POP: the probability it is positive.
- Read a profit diagram: the breakevens, the maximum profit (the credit grown to expiration), and the maximum loss. Closing early earns less, by roughly the extrinsic value left in the legs.

For understanding (you won't reproduce these)

- The NPV derivation and the formula tables for the eight composite contracts.

Lectures and Examples

Lecture:
CHEME-5660-L10b-Lecture-POP-POE-Delta-CompositeContracts-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Example 3:
CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

Review: The Greeks

For small changes in the share price, the volatility, the calendar time (years), and the rate, the premium $\mathcal{P}_\star$ of a call ($\star = c$) or put ($\star = p$) changes by:

$$\begin{aligned}\Delta\mathcal{P}_\star &\approx \Delta_\star\,\Delta S + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y \\ &= \underbrace{(\Delta_\star + \tfrac{1}{2}\,\Gamma_\star\,\Delta S)\Delta S}_{\text{share price}} + \underbrace{\nu_\star\,\Delta\sigma}_{\text{volatility}} + \underbrace{\Theta_\star\,\Delta t}_{\text{time}} + \underbrace{\rho_\star\,\Delta g_y}_{\text{rate}}\end{aligned}$$

- Delta and Gamma: the slope and curvature of the premium in S.
- Vega, Theta, and Rho: sensitivities to the volatility σ, the calendar time t, and the risk-free rate g_y, quoted as $\Theta_\star/365$ per day and $\nu_\star/100$, $\rho_\star/100$ per point.
- Hedge ratio: one contract covers 100 shares, so it is equivalent to $100 \cdot \Delta_\star$ shares.

Example 1: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Review: POE and POP

The POE is the probability of finishing in the money by more than 0.01 USD/share; the POP is the probability of a positive profit at expiration, beyond the breakeven $S^{\ddagger} = K \pm \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_\star$ (+ call, − put):

$$\text{POE} = \begin{cases} \mathbb{P}\left(S(T) > K + 0.01\right) & \text{call} \\ \mathbb{P}\left(S(T) < K - 0.01\right) & \text{put} \end{cases}$$

$$\text{POP} = \begin{cases} \mathbb{P}\left(S(T) > S^{\ddagger}\right) & \text{bought a call or sold a put} \\ \mathbb{P}\left(S(T) < S^{\ddagger}\right) & \text{sold a call or bought a put} \end{cases}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$; we simulate $S(T)$ under our belief $\mathbb{P}$. However, the POE $\sim |\Delta_\star|$, the Delta quoted by trading platforms under $\mathbb{Q}$.

Review: Computing POE and POP by Simulation

Initialize: given $S(0)$, K, the premium $\mathcal{P}_\star$ of the call or put, T (years), the drift μ and volatility σ under $\mathbb{P}$, g_y, and the number of samples M. Compute the breakeven $S^\ddagger$.

1. For each sample $j = 1, 2, \ldots, M$ **do**:
   a. Draw $Z_j \sim \mathcal{N}(0,1)$.
   b. Compute $S_j(T) \leftarrow S(0) \exp\left[(\mu - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_j\right]$.
2. Compute the fraction of samples at or below each share price x:
   $\hat{F}(x) \leftarrow \frac{1}{M} \cdot \#\left\{j : S_j(T) \le x\right\}$.

Review: Pseudocode for POE and POP

3. If the contract is a call, **then** POE $\leftarrow 1 - \hat{F}(K + 0.01)$. Otherwise, POE $\leftarrow \hat{F}(K - 0.01)$.
4. Compute the probability of profit:
   - If we bought a call or sold a put, **then** POP $\leftarrow 1 - \hat{F}(S^{\ddagger})$.
   - If we sold a call or bought a put, **then** POP $\leftarrow \hat{F}(S^{\ddagger})$.

With $\mu = g_y$ and σ set to the implied volatility, the same steps give the probabilities under $\mathbb{Q}$. These are the probabilities inside the market price, not our forecast.

Example 2: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Company Profile: Akuna Capital

Akuna Capital: an options market maker, headquartered in Chicago since 2011.

- Options market making: it posts prices to buy and sell, trading with its own money (a proprietary trading firm).
- Its own models: it designs its own trading systems and models.
- Teaching options: a free Options 101 course on payoff diagrams and the Greeks.

Connection: a market maker may take the other side of one or both legs of the spread we sell. The quoted Delta comes from its premiums, so $|\Delta_\star|$ approximates each leg's POE under $\mathbb{Q}$, not under our forecast $\mathbb{P}$.

Composite Contracts

A composite contract $\mathcal{C}$ combines long and short puts and calls on one underlying asset with one expiration date T. Its payoff per share at expiration is:

$$\hat{V}_{\mathcal{C}}(S(T)) = \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \overbrace{V_i(K_i, S(T))}^{\text{payoff of leg } i}}_{\text{sum of all legs}}$$

Here $S(T)$ is the share price at expiration, K_i the strike price of leg i, V_i its payoff, $(S(T) - K_i)^+$ for a call or $(K_i - S(T))^+$ for a put, n_i its number of contracts, and $\theta_i = -1$ if short or 1 if long.

The Net Present Value of a Composite Contract

A composite contract is a sum of legs, so its NPV is the sum of the legs' NPVs, weighted by $\theta_i n_i$:

$$\mathrm{NPV}_0(\mathcal{C}) = \sum_{i \in \mathcal{C}} \theta_i \, n_i \underbrace{\left(-\mathcal{P}_i + \overbrace{\mathcal{D}_{T,0}^{-1}(g_y) \cdot V_i(K_i, S(T))}^{\text{discounted payoff}} \right)}_{\text{NPV of buying leg } i} \qquad \text{weighted leg NPVs}$$

$$= -\underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \mathcal{P}_i}_{= \mathcal{P}_{\mathcal{C}}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, V_i(K_i, S(T))}_{= \hat{V}_{\mathcal{C}}(S(T))} \qquad \text{split the sum}$$

$$= -\mathcal{P}_{\mathcal{C}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \hat{V}_{\mathcal{C}}(S(T))$$

where $\mathcal{P}_i$ is the premium of leg i (USD/share), positive whether we buy or sell it (θ_i sets the direction), and $\mathcal{P}_{\mathcal{C}}$ is the composite premium.

Profit at Expiration

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ writes the NPV as a profit at expiration discounted to today:

$$\mathrm{NPV}_0(\mathcal{C}) = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\left(\hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}\right)}_{= \hat{P}_{\mathcal{C}}(S(T))} \qquad \text{factor out } \mathcal{D}_{T,0}^{-1}(g_y)$$

$$= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\hat{P}_{\mathcal{C}}(S(T))}_{\text{profit at expiration}}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor, and:

$$\boxed{\hat{P}_{\mathcal{C}}(S(T)) = \hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}}$$

is the profit at expiration (USD/share). $\mathcal{P}_{\mathcal{C}} > 0$ opens for a debit; $\mathcal{P}_{\mathcal{C}} < 0$ opens for a credit of $-\mathcal{P}_{\mathcal{C}}$.

Probability of Profit and Probability of Exercise

The NPV and the profit at expiration have the same sign, so with $\mathbb{P}$ our belief, the probability of profit is:

$$\boxed{\text{POP}_{\mathcal{C}} = \mathbb{P}\Big(\text{NPV}_0(\mathcal{C}) > 0\Big) = \mathbb{P}\Big(\hat{P}_{\mathcal{C}}(S(T)) > 0\Big)}$$

Each leg has its own probability of exercise:

$$\text{POE}_i = \begin{cases} \mathbb{P}\left(S(T) > K_i + 0.01\right) & \text{if leg } i \text{ is a call} \\ \mathbb{P}\left(S(T) < K_i - 0.01\right) & \text{if leg } i \text{ is a put} \end{cases}$$

Common Composite Contracts

By choosing which puts and calls we sell and buy, we can build trades that are bullish, bearish, or neutral:

- Naked: a short contract with no long contract to cap its loss.
- Credit: each contract below opens for a credit, so $\mathcal{P}_{\mathcal{C}} < 0$, and each profit at expiration starts with the credit grown to expiration, $-\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}$.

Common Composite Contracts: Outlook

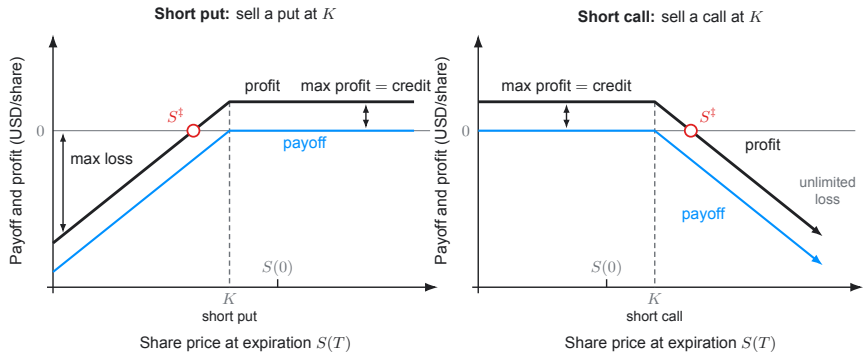
The eight structures, by side:

Composite contract	Put side	Call side	Outlook
Short put	naked short put	none	neutral to bullish
Short call	none	naked short call	neutral to bearish
Put credit spread	put credit spread	none	neutral to bullish
Call credit spread	none	call credit spread	neutral to bearish
Short straddle	naked short put	naked short call at the same strike	neutral
Short strangle	naked short put	naked short call at a higher strike	neutral
Protected strangle	naked short put	call credit spread	neutral, leaning bullish
Iron condor	put credit spread	call credit spread	neutral

Strikes are numbered low to high, $\mathcal{P}_i$ is the premium of the leg at K_i. Figures: payoff (blue), profit (black), and $S(0)$, the share price at entry.

Short Puts and Short Calls

Selling one put, a short put, or one call, a short call, is the building block of every structure below:



Each keeps the whole premium if it expires out of the money.

Short Puts and Short Calls: Profit at Expiration

With premiums $\mathcal{P}_p$ (put) and $\mathcal{P}_c$ (call) at strike K, the profits at expiration, and the breakevens $S^\ddagger$, are:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p - (K - S(T))^+ \qquad \text{short put}$$

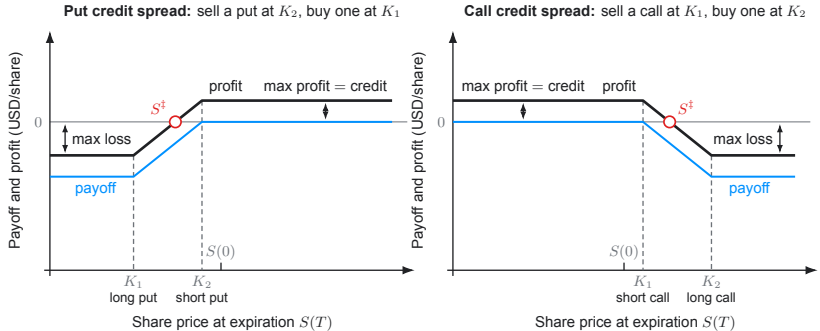
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c - (S(T) - K)^+ \qquad \text{short call}$$

	Short put	Short call
Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ for $S(T) \geq K$	$\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$ for $S(T) \leq K$
Breakeven	$S^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$	$S^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$
Maximum loss	$K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ at $S(T) = 0$	unlimited

The short put's loss is bounded ($S(T) \geq 0$); the short call's is not.

Credit Spreads

A credit spread sells the contract closer to the share price and buys one farther away, same type and expiration (a bull put or bear call spread):



The short leg collects the credit; the long leg caps the loss.

Put Credit Spread

A long put at K_1 and a short put at K_2: the credit grown to expiration, minus and plus the leg payoffs:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1) - (K_2 - S(T))^+ + (K_1 - S(T))^+$$

Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \geq K_2$
Breakeven	$S^\ddagger = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$
Maximum loss	$(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \leq K_1$

Example 3: CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

Call Credit Spread

A short call at K_1 and a long call at K_2 give the profit at expiration:

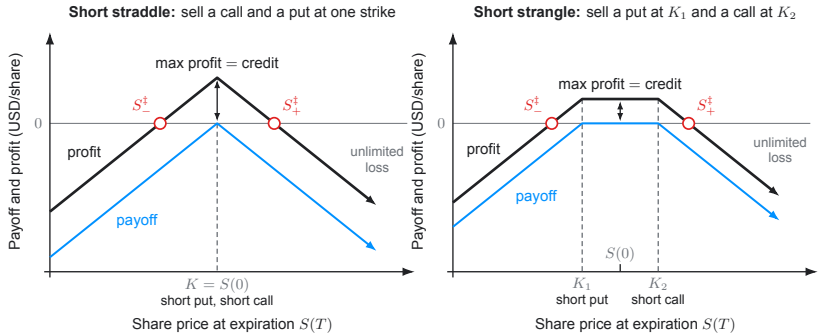
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2) - (S(T) - K_1)^+ + (S(T) - K_2)^+$$

Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \leq K_1$
Breakeven	$S^\ddagger = K_1 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$
Maximum loss	$(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$

Beyond the long strike, the long leg gains what the short leg loses, so the loss stops at the strike width minus the credit.

Short Straddles and Strangles

A straddle: a put and a call at one strike K. A strangle: a put at K_1 below a call at K_2. Same expiration:



Sold, each profits only between its breakevens: POP $= \mathbb{P}(S_-^\ddagger < S(T) < S_+^\ddagger)$.

Short Straddle

Selling a put and a call at the strike K gives the profit at expiration:

$$\widehat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c) - (K - S(T))^+ - (S(T) - K)^+$$

Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = K$
Lower breakeven	$S_-^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$
Maximum loss below $S_-^\ddagger$	$K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = 0$
Upper breakeven	$S_+^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$
Maximum loss above $S_+^\ddagger$	unlimited

The straddle collects more credit, but the strangle keeps its credit over a wider range of share prices.

Short Strangle

Selling a put at K_1 and a call at K_2 gives the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2) - (K_1 - S(T))^+ - (S(T) - K_2)^+$$

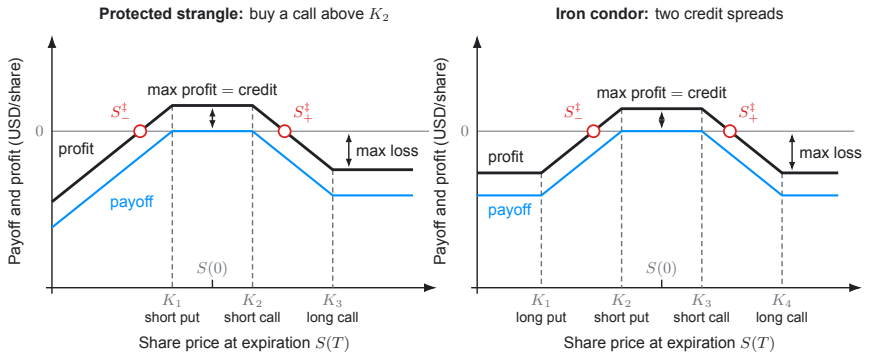
Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ for $K_1 \leq S(T) \leq K_2$
Lower breakeven	$S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$
Maximum loss below $S_-^\ddagger$	$K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ at $S(T) = 0$
Upper breakeven	$S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$
Maximum loss above $S_+^\ddagger$	unlimited

Buying both legs ($\theta_i = 1$) flips either diagram: the loss is capped at the premium paid, and a large move either way pays.

Review on your own: CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

Protected Strangles and Iron Condors

A protected strangle (jade lizard) adds a long call at K_3 to a short strangle; an iron condor also adds a long put:



Each long leg caps the loss on its side but lowers the credit.

Protected Strangle

A short put at K_1, a short call at K_2, and a long call at K_3 give the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) \qquad \text{protected strangle}$$
$$- (K_1 - S(T))^+ - (S(T) - K_2)^+ + (S(T) - K_3)^+$$

Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $K_1 \leq S(T) \leq K_2$
Lower breakeven	$S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$
Maximum loss below $S_-^\ddagger$	$K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ at $S(T) = 0$
Upper breakeven	$S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)^a$
Maximum loss above $S_+^\ddagger$	$(K_3 - K_2) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $S(T) \geq K_3{}^b$

[a] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) \leq K_3 - K_2$; otherwise none. [b] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) < K_3 - K_2$; otherwise none.

If the credit grown to expiration is at least $K_3 - K_2$, the call side cannot lose.

Iron Condor

A long put at K_1, a short put at K_2, a short call at K_3, and a long call at K_4 give the profit at expiration:

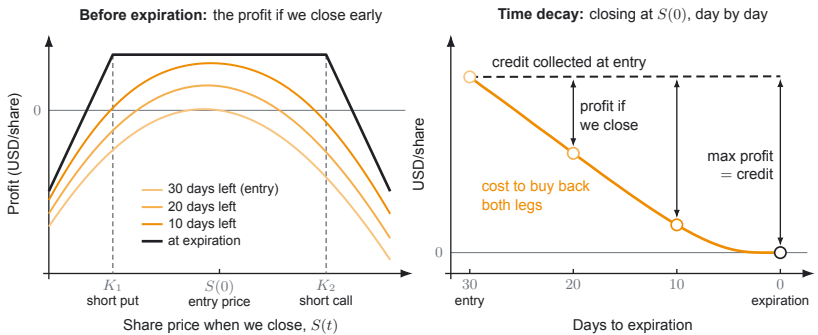
$$\begin{aligned}\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4) \\ + (K_1 - S(T))^+ - (K_2 - S(T))^+ - (S(T) - K_3)^+ + (S(T) - K_4)^+\end{aligned}$$

Maximum profit	$\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $K_2 \le S(T) \le K_3$
Lower breakeven	$S_-^\ddagger = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$
Maximum loss below $S_-^\ddagger$	$(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \le K_1$
Upper breakeven	$S_+^\ddagger = K_3 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$
Maximum loss above $S_+^\ddagger$	$(K_4 - K_3) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \ge K_4$

At any share price at expiration, at most one side can lose, so the maximum loss is the wider strike width minus the credit.

Why Wait to Expiration?

Our diagrams are drawn at expiration. Closing a short leg early means buying it back: intrinsic plus extrinsic value $E_V(t)$. A short strangle (put $K_1 = 90$, call $K_2 = 110$, $S(0) = 100$ USD/share, 30 days, CRR tree, $\sigma = 0.40$, $g_y = 4\%$):



Why Wait to Expiration? What Do We See?

- Before expiration (left): the buyback price still includes extrinsic value, so earlier curves lie below the profit at expiration.
- At the entry share price (right): both legs are out of the money, so closing at once costs exactly the credit and earns nothing.
- Time decay: as the extrinsic value decays, the profit if we close at $S(0)$ rises to the whole credit.

The diagrams show the most each contract can earn; an early exit earns less, by roughly the extrinsic value left in the legs.

Summary

Key takeaways

- POE, POP, and Delta: both probabilities come from simulated share prices, and the size of Delta approximates the POE under $\mathbb{Q}$.
- The NPV of a composite contract is a sum over its legs: the POP is the probability that the profit at expiration is positive.
- Common composite contracts can be bullish, bearish, or neutral: the profit diagram gives the outlook, breakevens, and maximum loss; closing early earns less.